

A USER FRIENDLY DOCUMENTATION FOR LILY'S PAD



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Lily's Pad

1. Project presentation

1.1 Goal

Lily's Pad is a pedagogic device helping its users learn sign language (both British Sign Language and French Sign Language) in an intuitive and playful way. The device helps discover the alphabet and the digits for French Sign Language (FSL) and English Sign Language (ESL) through several interactive modes (translation, alphabet, guess the sign/letter).

Some typical users for Lily's Pad are :

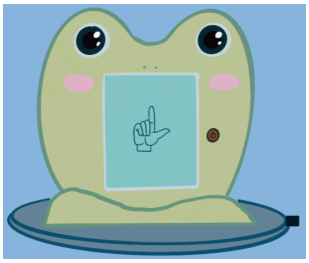
- Children learning sign language (aged 5+)
- Teachers and teaching staff
- Deaf or partially deaf people's families
- Any person wanting to learn the basics of sign language

1.2 Concept

The device is shaped after a frog sitting on a lilypad, which is where the keyboard is placed.

A main LCD screen displays the words, highlighting the letter for the current sign displayed, whereas the two smaller auxiliary screens are used to display the frog's animated eyes in order to make Lily's Pad more lively and attractive to kids.

Users shall interact with the device thanks to the keyboard, as well as the rotary encoder and a slide switch to pick the language.



2. Specifications

2.1 Main features

- Displays the alphabet in sign language
- Translates the inputted letters into FSL/BSL signs.
- Educational game modes
- Compatible with French Sign Language and British Sign Language
- Animated eyes
- Power supply via USB Type-C

2.2 Operating modes

Translation Mode

The user inputs a word, letter by letter, using the keyboard. Each letter is displayed on the main LCD Screen under its sign language equivalent.

Game Mode

Two educational game modes are available:

Guess the sign

A letter is displayed, the user must pick the sign corresponding to the displayed letter, using the rotary encoder.

Guess the letter

A picture of a sign is displayed, the user must find the corresponding letter, using the keyboard.

Alphabet Mode

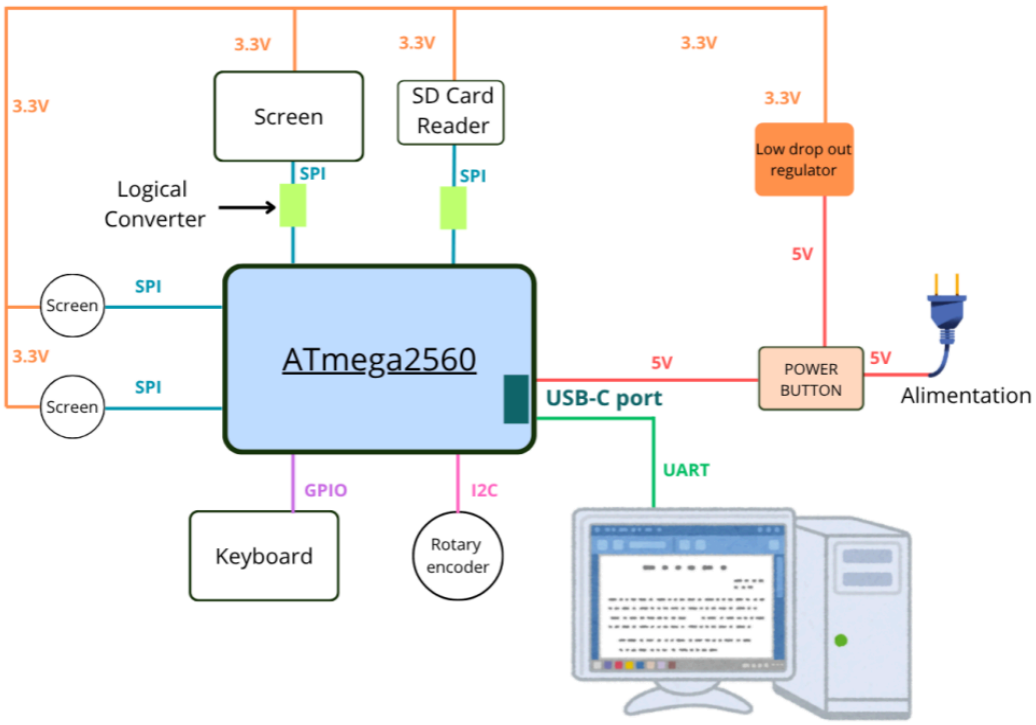
The user uses the rotary encoder to scroll through all the signs of the selected language.

3. Hardware

3.1 Main Components

Component	Function
Main LCD Screen	Displays words, letters and/or signs
Auxiliary LCD screens	Animated eyes
Keyboard	User input for letters
Rotary encoder	User input for signs
Rotary encoder's pushbutton	Input validation
i2C I/O expander	connects Rotary encoder EC11
USB Type-C port	Power supply and UART canal
Power button	ON/OFF
Slide switch	Selection FSL / BSL
Microcontroller	System control

3.2 Functional diagram



4. Design and Integration of casing

4.1 Overview of casings

The device will consist of two parts : an upper casing and a lower casing, both conceived using the program Fusion and 3D printed.

4.1.1 Upper Frog shaped casing

This casing includes:

- The LCD main screen
- The two auxiliary LCD screens for the frog's eyes
- The main PCB (MCU + screens pins + pins to connect the keyboard)

4.1.2 Lower Water Lily shaped casing

This casing includes:

- The keyboard as well as the rotary encoder
- Opening for the mounting of the power button
- The keyboard PCB (keyboard + rotary encoder + slide switch + pins to connect to main PCB)



Lily's PAD
3D Exploded Assembly View

LILY'S PAD


BONJOUR
Lily's PAD
1 2 3 4 5 6 7 8 9 0
Q W E R T Y U I O
A S D F G H J K L
Z X C V B N M

5. Software architecture

5.1. Program's main structure

The FIRMWARE is divided in different parts:

- Display drivers (main screen - ILI9488 & eyes - GC9A01)
- Keyboard driver
- Rotary encoder driver
- SD card driver
- Files discovery on the SD card (FAT32)
- Main code of the game
- Eyes animation

The firmware can be updated via the USB Type-C connection to a computer

To do this you need to have avr-dude installed on your system.

5.2 How to update Lily's pad ?

- Download update.sh in the latest releases section on our Git hub
- Run: chmod +x update.sh
- Run: ./update.sh <DEVICE PORT> <OPTIONAL: THEME>

  DEVICE PORT:

  Path to the port used by the USB jack that the Lily's Pad is connected to.
  Example: "/dev/cu.usbserial-1430"

  OPTIONAL THEMES:

  run " ./update.sh -h " and the list will appear with the themes available

6. User guide

6.1 Powering up

1. Plug the device with the USB-C cable.
2. Press the power button.
3. Wait for the welcome screen to appear, then the menu.

6.2 Waking up Lily's Pad

When Lily'sPad goes to sleep (after a few tens of seconds), wake her up by touching the keyboard or turn/click Lily's Flower until she wakes up.

6.3 Language selection

Use the slide switch :

- Position 1 (Left) : English sign language. 'BSL'
- Position 2 (Right) : French sign language. 'LSF'

6.4 Using translation mode

1. Select the translation mode.
2. Enter letters using the keyboard. (14 letters max)
3. Observe the corresponding representation in sign language.

6.5 Using game mode

1. Select game mode.
2. Choose one of the two available games (Guess the sign/letter).
3. Follow the instructions on the screen.
4. Play the game.

6.6 Using alphabet mode

1. Select the alphabet mode
2. Rotate the encoder to cycle through the different signs.

7. Trouble Shooting Guide

If something happens to your Lily'sPAD, don't panic ! Here is a little list what you can do:

1. White main screen : it sounds like a connection problem, open up the back, two small screws (Allen, 2.5). and check that all the cables are firmly connected.

   Check also the SD card, top left corner, and a small card. check that it's firmly in the reader. Then restart, using the RESET button on the back of the device..

2. All screens are black : Check the power supply, i.e. the USB cable, is it working and connected properly, or is the POWER button at the side of Lily's Pad turned on. If not, turn it on, silly you !

3. Screen is blocked on an image/text and the keyboard doesn't seem to work: You have two options, turn the POWER button off then back on or with a pen press the RESET button on the back (yellow button) and release

4. If you can not solve the problem call Louise our super technical advisor et the number 06 42 42 42 42.

8. Conclusion

Lily's Pad successfully combines hardware engineering, software engineering and product design in order to provide a comprehensible introduction to sign language.

Its fun frog shape and the educational learning modes make Lily's Pad an astounding tool adapted for kids, teachers and families that wish to learn more about FSL/ESL in an engaging and amusing manner.